$$\frac{\mathrm{d}\mathbf{\Gamma}(\boldsymbol{\gamma}(t))}{\mathrm{d}t} = \mathfrak{L}\,\mathbf{\Gamma}(\boldsymbol{\gamma}(t)) \quad \mathbf{\Gamma}(\boldsymbol{\gamma}(0)) \equiv \mathbf{\Gamma}(\boldsymbol{\gamma}_0) \quad \mathfrak{L} \equiv iL \coloneqq \{\ldots, \mathcal{H}\} \quad \boldsymbol{\gamma} \in X = \mathbb{R}^{6M^{FG}}$$

$$\mathbf{A}(\boldsymbol{\gamma}) = \left\{ A_I \in L^2\left(X, \mathbb{P}\right), \text{ where } I = 1, \ldots, M^{CG} \text{ and } M^{CG} \leq M^{FG} \right\} \in V = \mathbb{R}^{M^{CG}}$$

$$\mathcal{P}_{\mathbf{a}} F(\boldsymbol{\gamma}) = \frac{1}{\Omega(\mathbf{a})} \int_X F(\boldsymbol{\gamma}')\, \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \rho^{eq}(\boldsymbol{\gamma}') \mathrm{d}\boldsymbol{\gamma}', \qquad \Omega(\mathbf{a}) = \int_X \delta\left(\mathbf{\Gamma}(\boldsymbol{\gamma}') - \mathbf{a}\right) \mathbb{P}(\mathrm{d}\boldsymbol{\gamma}')$$

$$\mathcal{P}_{\mathbf{a}}^{+} = \mathcal{P}_{\mathbf{a}}$$